Materials Science 1 – Diffusion in Solids
Part 1

General learning outcomes of diffusion part

1. Describe the mechanisms of diffusion in solid materials;

2. Calculate diffusion flux using Fick's first and second law;

3. Distinguish between steady-state and non-steady state diffusion;

4. List the principal factors that affect diffusion in solids.

Learning objectives and learning outcomes of this lecture

1. Describe the main mechanisms of diffusion in solids at atomic level;

2. Outline Fick's first law;

3. List and explain the main mechanism of diffusion in solids;

4. Apply Fick's first law for steady-state diffusion to calculate the diffusion flux in solids.

Poll – Answers in WeChat

Q1) Diffusion is the result of which of the following process?

a) Random motion of particles;
b) Concentration gradient;
c) Kinetic energy of particles;
d) All of the above.

Poll – Answer

Q1) Diffusion is the result of which of the following process?

D) All of the above

Diffusing particles (which can be in all three phases) possess kinetic energy and are constantly under random motion. However, the rate of this movement is proportional to the number of particles available and hence the net direction of diffusion is from areas with higher concentration to areas with lower concentration.

Diffusion

Diffusion of a dye in a liquid, the blue dye diffuses in water occupying a larger space in the liquid.

Mass transport by atomic motion from a region of high concentration to a region of low concentration. It can occur in gases, liquids and solids.

Diffusion

The mechanism of diffusion depends on the state of matter.

- **Gases:** random Brownian motion;

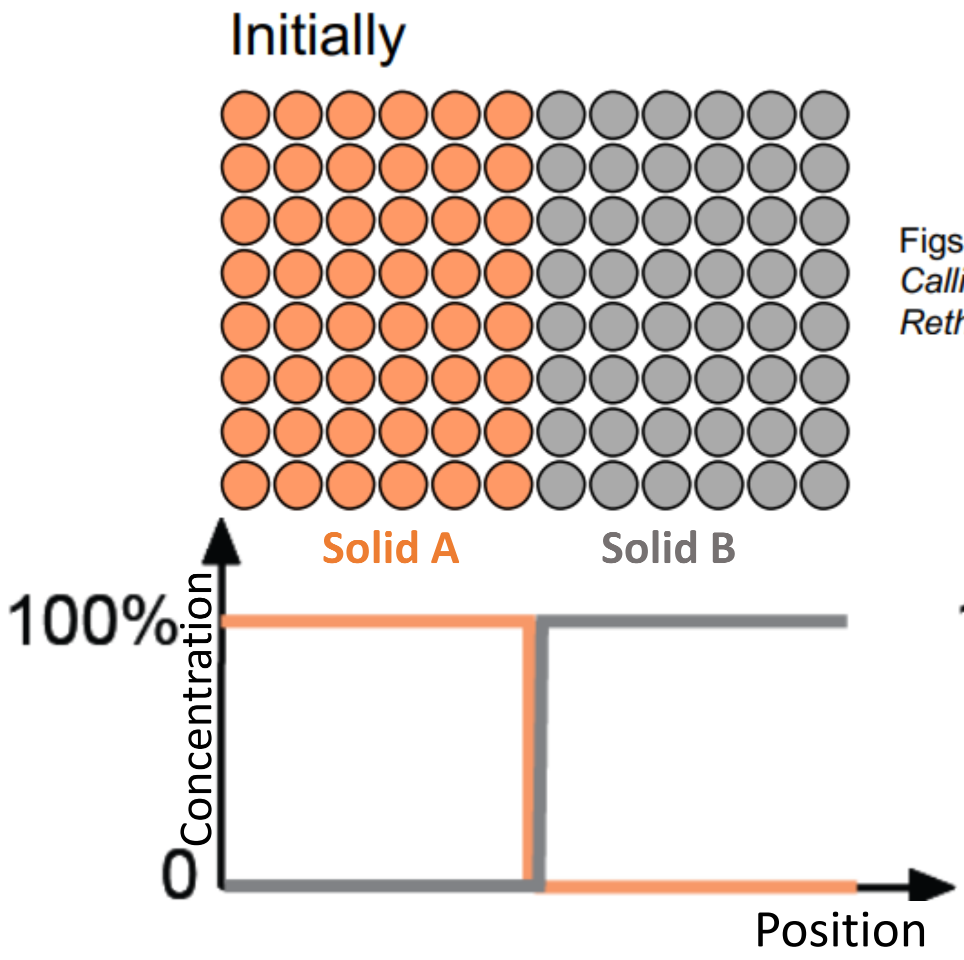
- **Liquids:** random Brownian motion;

- **Solids:**

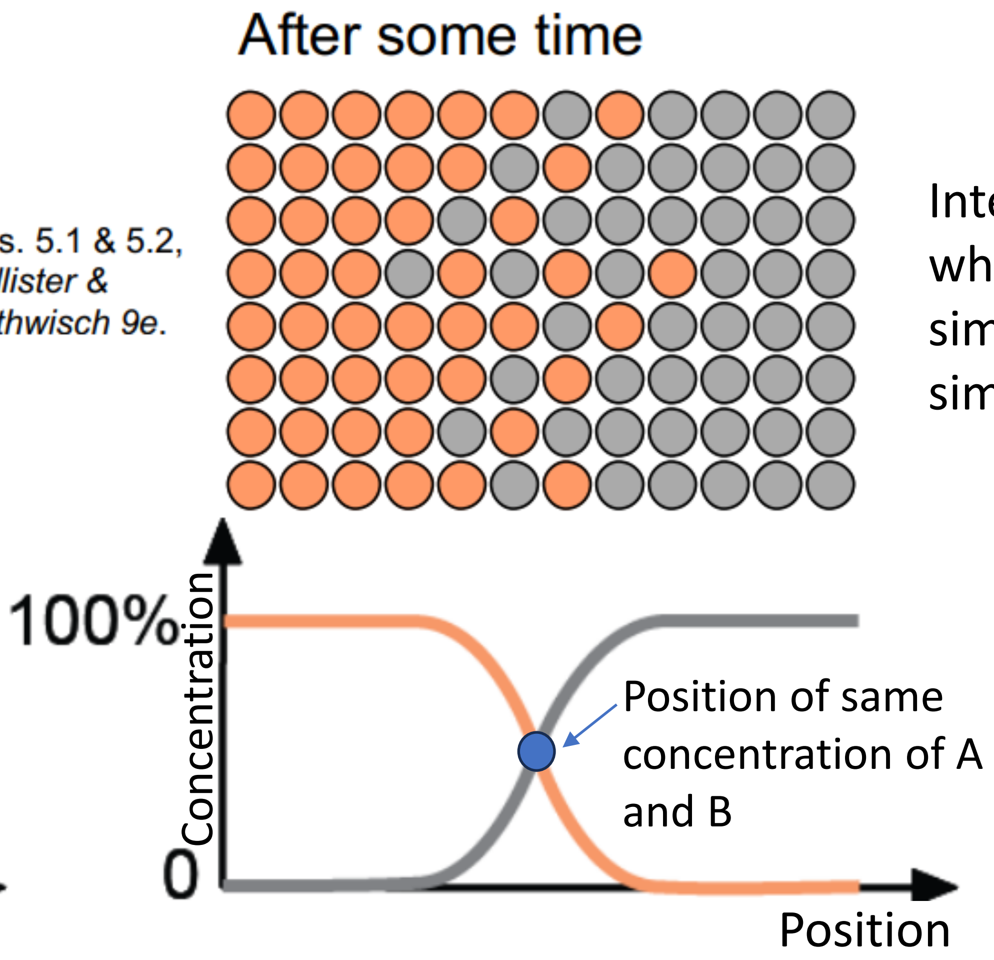
i) Vacancy diffusion;
ii) Interstitial diffusion.

Diffusion in solids

1. <u>Interdiffusion:</u> In an alloy, atoms tend to migrate from regions of high concentration to regions of low concentration.



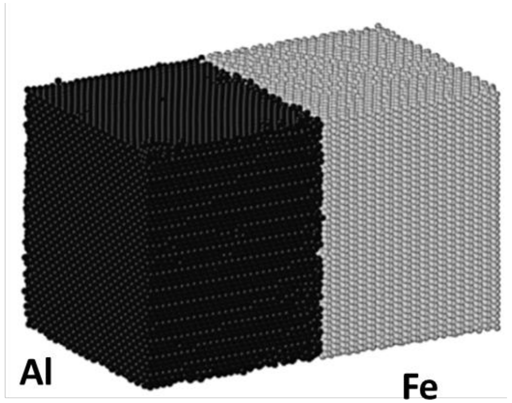
Figs. 5.1 & 5.2, Callister & Rethwisch 9e.



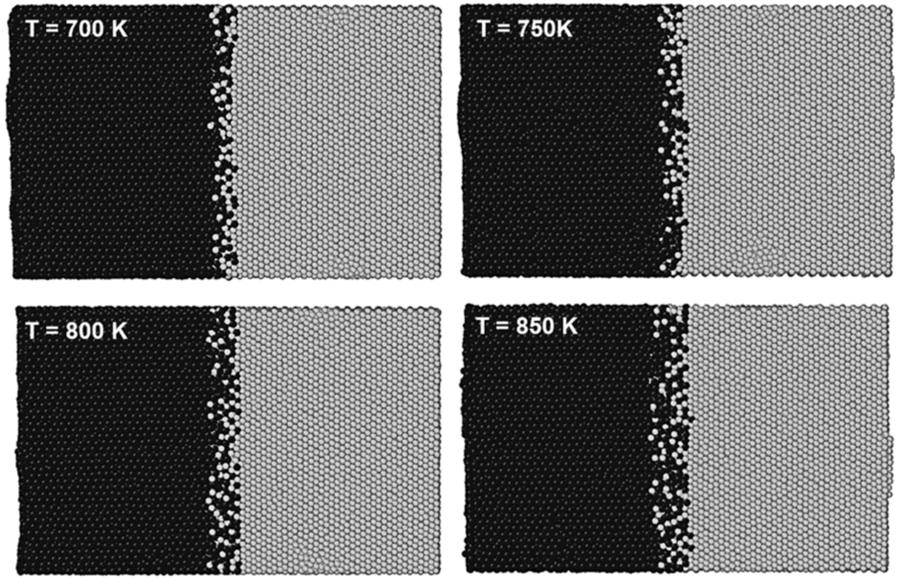
Interdiffusion is facilitated when solid A and B have similar atomic radius and similar crystal structures.

Diffusion in solids simulation

Simulation of interdiffusion across an interface between two solids: Al (black) and Fe (grey):



Snapshots of the interface after 6ns at different temperatures



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- Fe atoms penetrated the Al lattice deeper than Al into Fe lattice due to the lower atomic radius.

- The high temperatures close to the melting point of Al resulted in weak bonding between the Al atoms. This caused the formation of vacancies in the Al lattice that helped Fe atoms to migrate in.

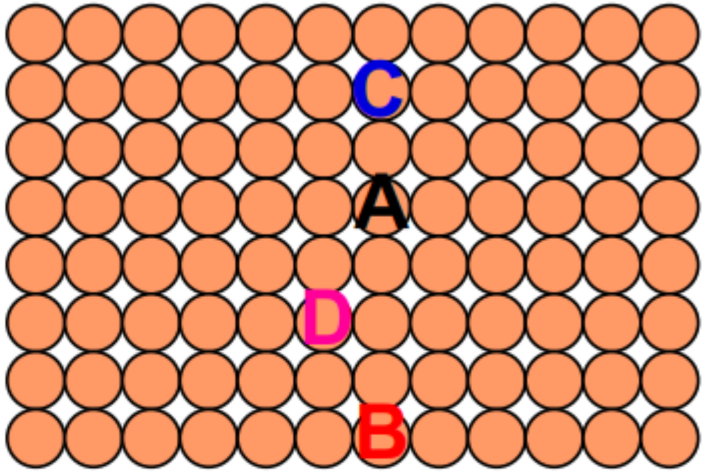
The rate of substitutional diffusion depends on:

i) vacancy concentration;
ii) frequency of jumping;
iii) temperature.

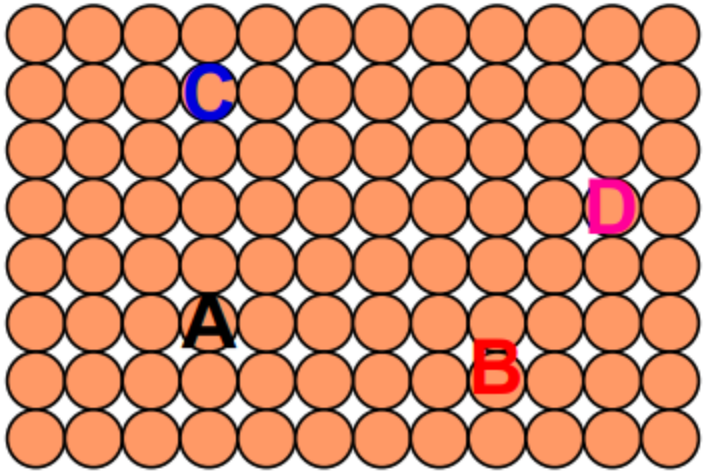
Diffusion in solids

2. <u>Self-diffusion:</u> In a mono-elemental solid, atoms migrate to other sites. This process does not lead to any observable compositional changes.

Label some atoms



After some time



Diffusion in solids - Mechanisms

The atoms in solid materials are in constant motion. For an atom to change position, two main conditions must be met:

(1) there must be an empty adjacent site;

(2) the atom must have sufficient energy to break bonds with its neighbour atoms and then cause some lattice distortion during the displacement.

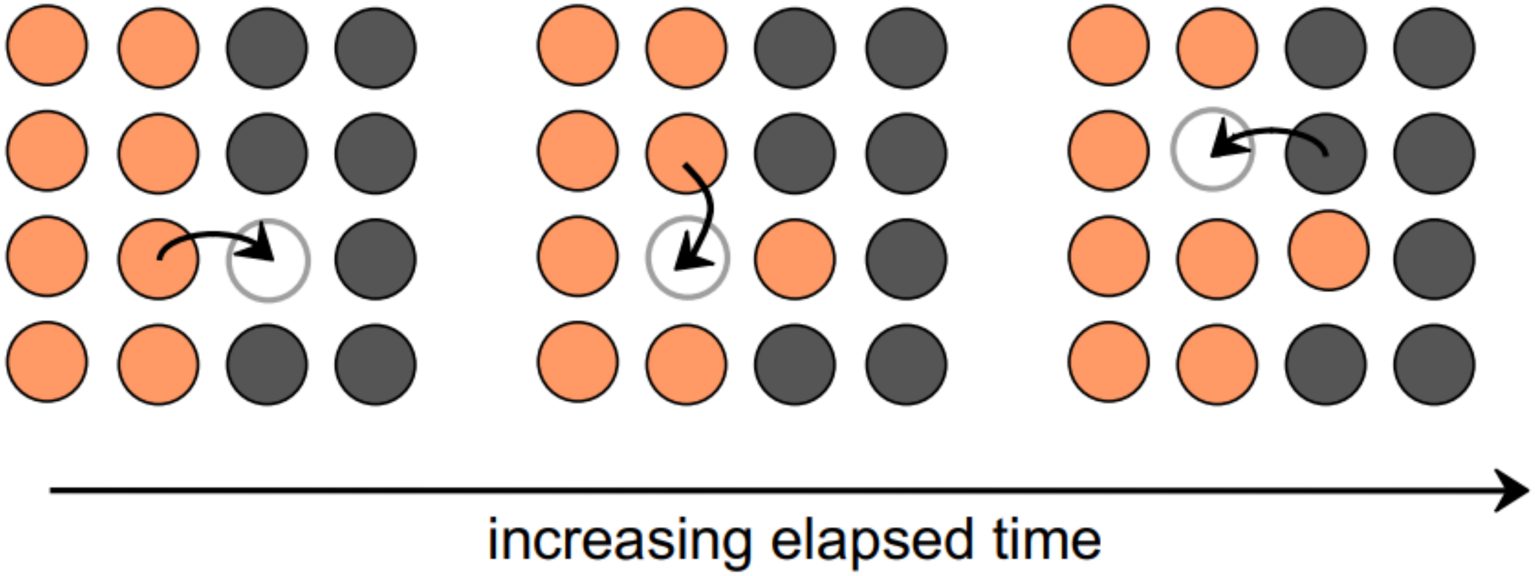
Diffusion in solids - Mechanisms

Vacancy diffusion:

- Atoms exchange with vacancies (diffusion of atoms in one direction corresponds to the motion of vacancies in the opposite direction);
- Applies to substitutional impurities atoms;
- Diffusion rate depends on:

i) number of vacancies;

ii) activation energy to exchange.



Both self-diffusion and interdiffusion occur by this mechanism; for the latter, the impurity atoms must substitute for host atoms.

Diffusion in solids - Mechanisms

Interstitial diffusion: smaller atoms can diffuse in the interstitial spaces between atoms.

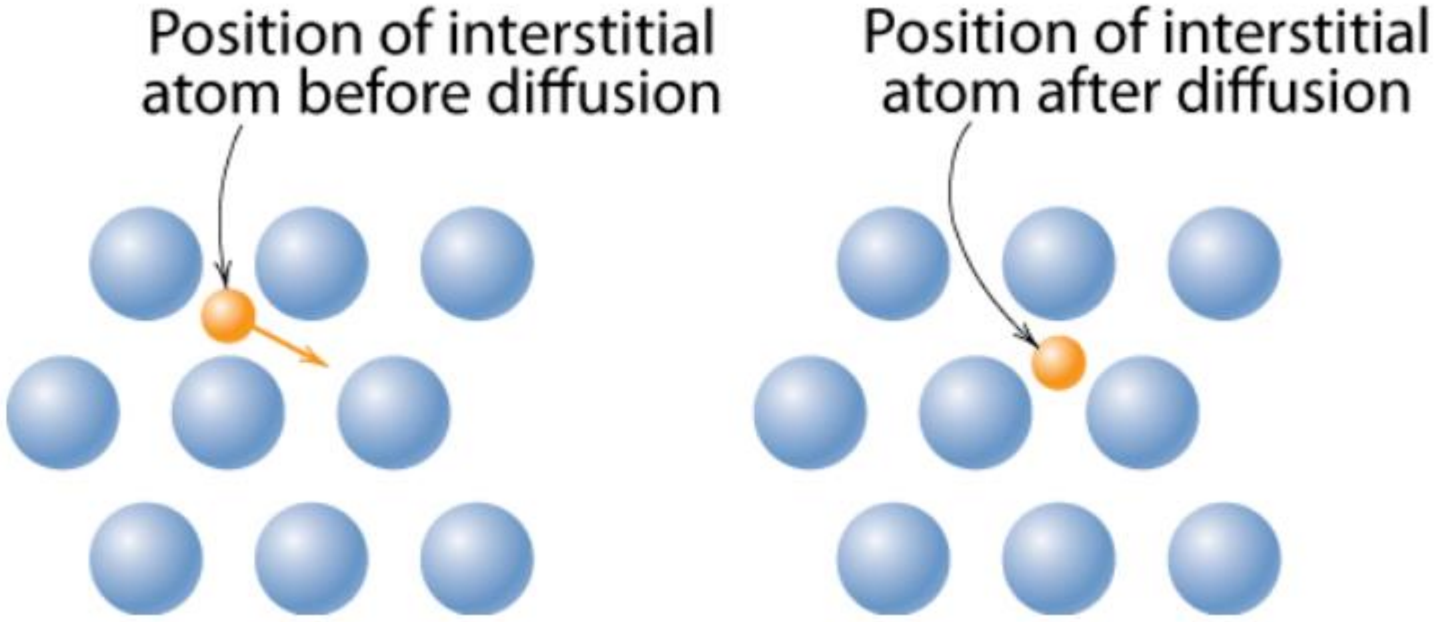


Fig. 5.3 (b), *Callister & Rethwisch 9e.*

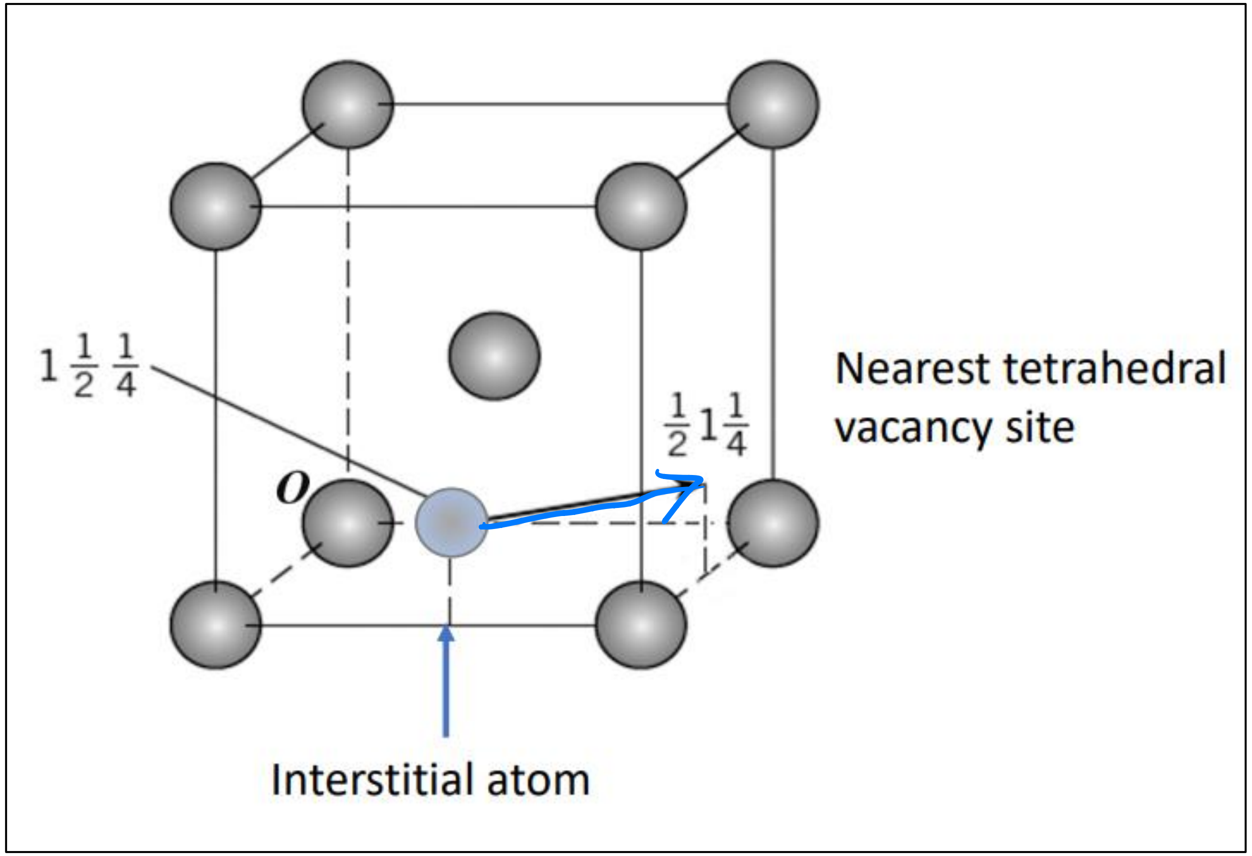
In most metal alloys, interstitial diffusion is faster than vacancy diffusion. Why?

i) Interstitial atoms are smaller and thus more mobile.

ii) There are more empty interstitial positions than vacancies, leading to higher probability of interstitial atomic movement than vacancy diffusion.

Diffusion of atoms between different positions in a crystal structure

Consider an interstitial atom initially placed in the $1\frac{1}{2}\frac{1}{4}$ position of a BCC structure.



If the atom diffuses to a vacancy site in position $\frac{1}{2}1\frac{1}{4}$, what is the vector that identifies the diffusion direction?

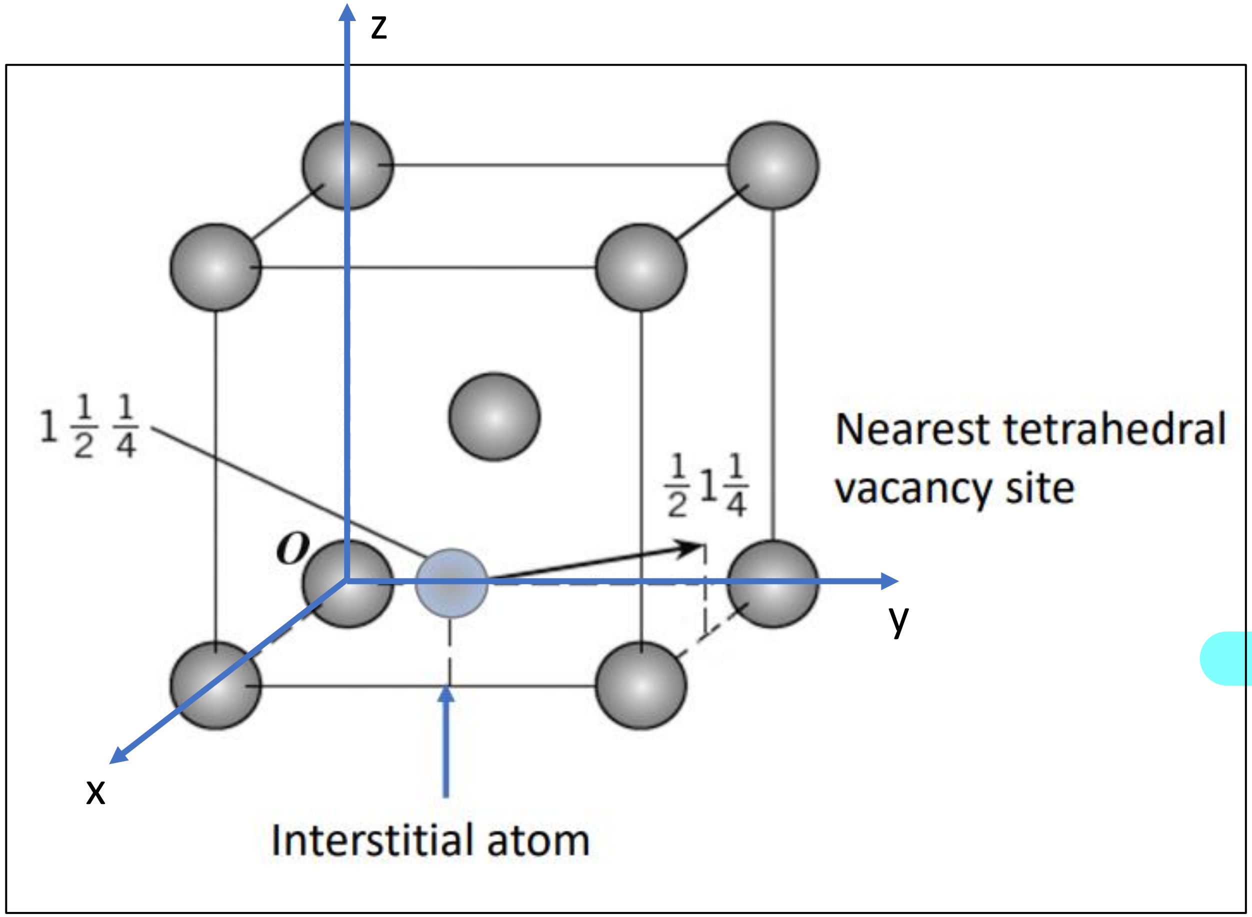
vector 头 - 尾

$\frac{1}{2}\,1\,\frac{1}{4} - 1\,\frac{1}{2}\,\frac{1}{4}$

Diffusion of atoms between different positions in a crystal structure

Consider an interstitial atom initially placed in the $1\frac{1}{2}\frac{1}{4}$ position of a BCC structure.

What is the direction of the vector if the atom diffuses to a vacancy site in position $\frac{1}{2}1\frac{1}{4}$?



Algebrically, this can be identified by the difference between the two vectors:

$$\left[\frac{1}{2}1\frac{1}{4}\right] - \left[1\frac{1}{2}\frac{1}{4}\right] = [-\frac{1}{2}\frac{1}{2}0]$$

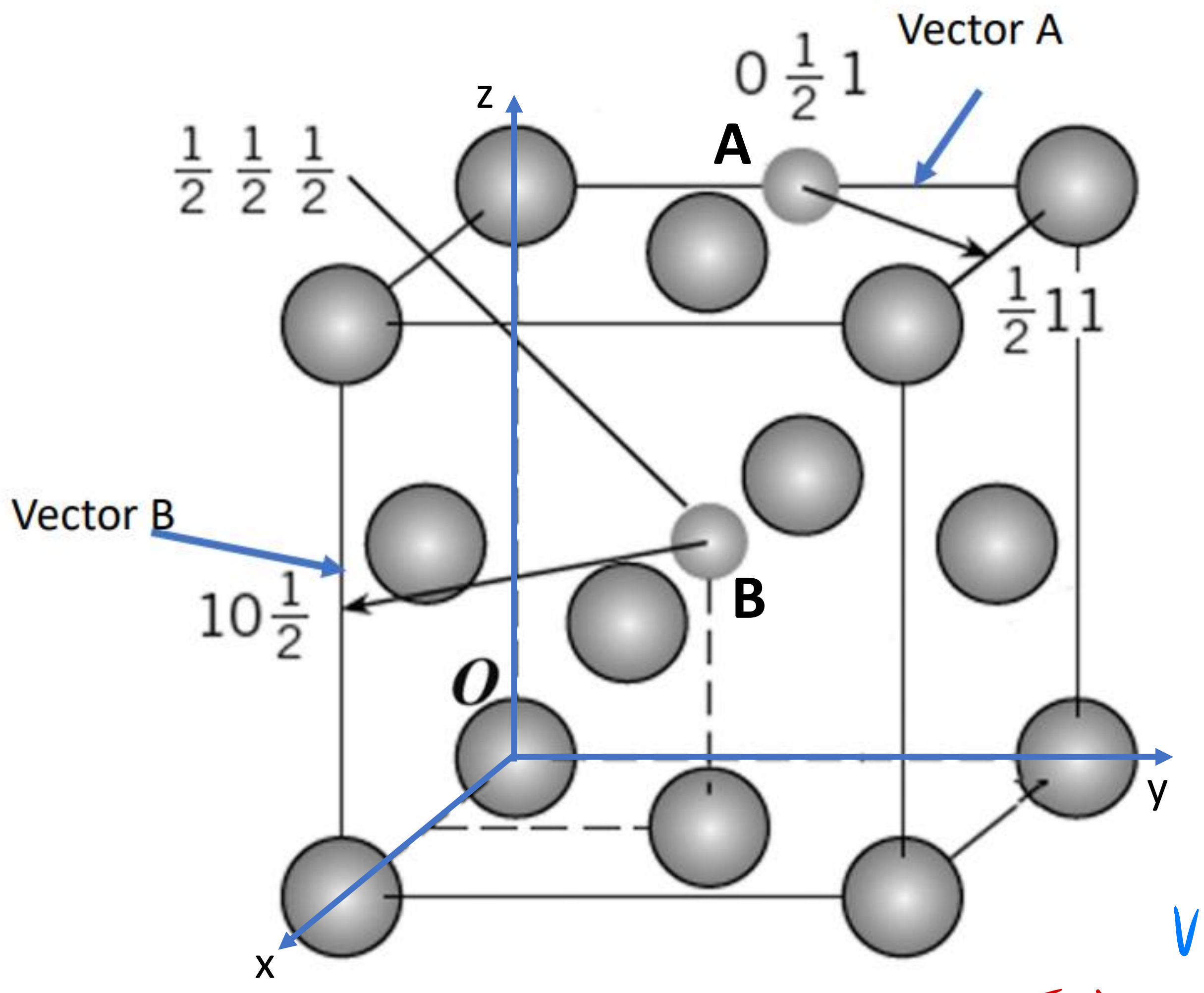
This means that the atom has moved along x of $-\frac{1}{2}$, along y of $\frac{1}{2}$ and along z of 0.

☆ 不要忘记化简

Reducing the residual by x 2, we have the equivalent vector direction: $[\bar{1}10]$.

Diffusion of atoms between different positions in a crystal structure

Example 1: Diffusion of interstitial atoms in an FCC structure.



What would be the direction of the vectors A and B shown in the figure, if the atoms diffuse to a vacancy site with the coordinates depicted?

vectorA $[\frac{1}{2}\ 1\ 1] - [0\ \frac{1}{2}\ 1]$

$= [\frac{1}{2}\ \frac{1}{2}\ 0] = [1\ 1\ 0]$

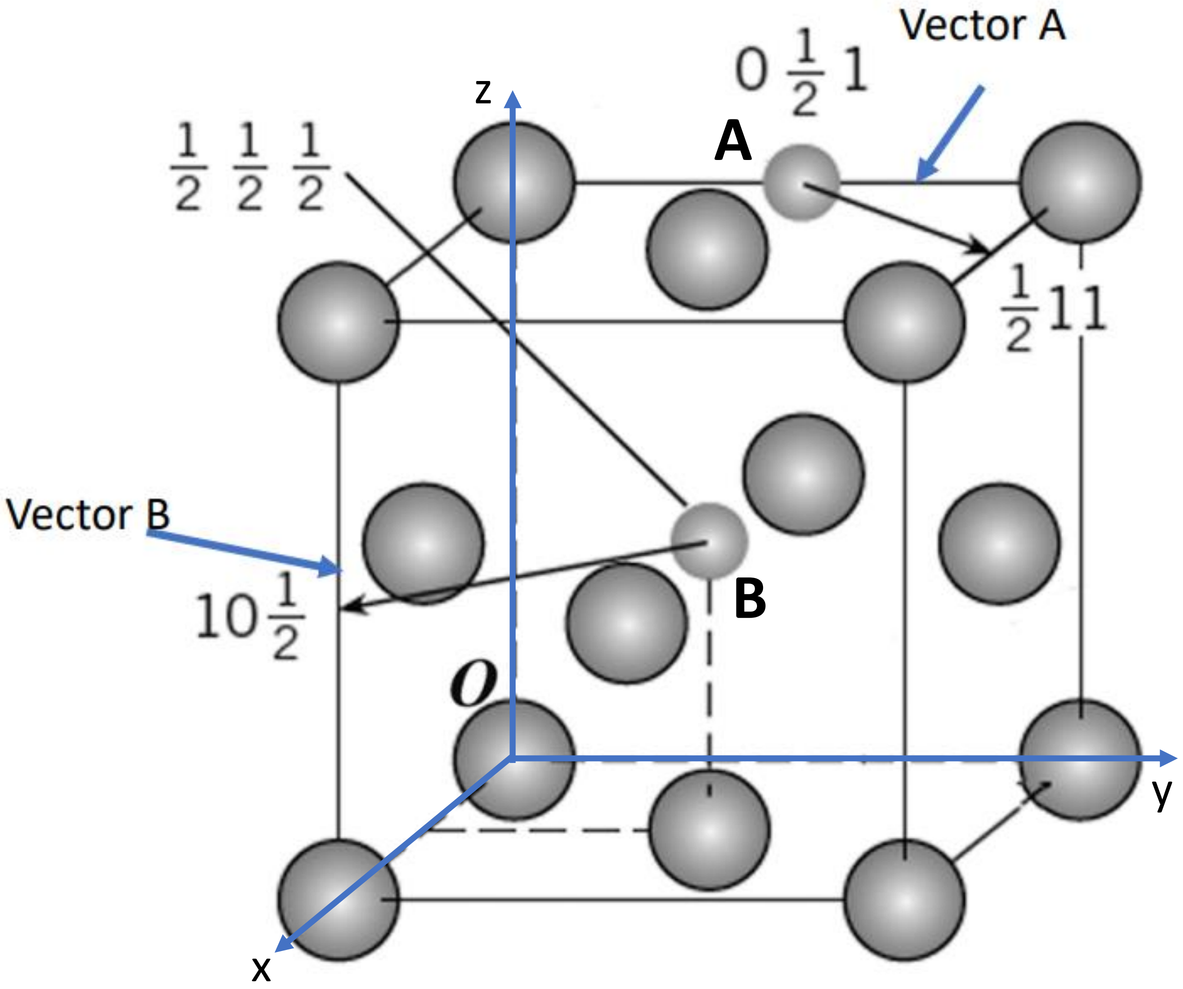
vector B $[1\ 0\ \frac{1}{2}] - [\frac{1}{2}\ \frac{1}{2}\ \frac{1}{2}]$

$= [\frac{1}{2}\ -\frac{1}{2},\ 0] = [1\ \bar{1}\ 0]$

不用逗号

Diffusion of atoms between different positions in a crystal structure

Example 1 solution: Diffusion of interstitial atoms in an FCC structure.



Vector A

x coordinate: $\frac{1}{2} - 0 = \frac{1}{2}$

y coordinate: $1 - \frac{1}{2} = \frac{1}{2}$

z coordinate: $1 - 1 = 0$

Reducing the fraction by x 2

Direction: [110]

规范

Vector B

x coordinate: $1 - \frac{1}{2} = \frac{1}{2}$

y coordinate: $0 - \frac{1}{2} = -\frac{1}{2}$

z coordinate: $\frac{1}{2} - \frac{1}{2} = 0$

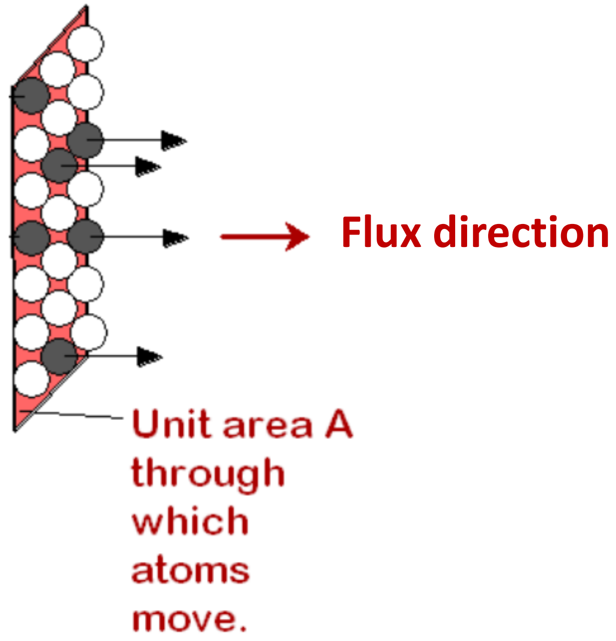
Reducing the fraction by x 2

Direction: $[1\bar{1}0]$

Parameters characterizing the diffusion process

Diffusion is a time dependent process: the mass transport of a diffusing species within a medium is a function of time;

Diffusion rate: is quantified by the rate of mass transfer, frequently expressed by the diffusion flux (J), defined as the mass (or number of atoms) M diffusing through and perpendicular to a unit cross-sectional area (A) per unit of time (t):



$$J = \frac{M}{At}$$

The units for J are kilograms or atoms per meter squared per second (kg/(m^2·s) or atoms/(m^2·s)).

Poll – Answers in WeChat

Q1) What would diffuse faster through a porous solid?

a) Gas molecules
b) Liquid molecules.

Poll – Answer

Q1) What would diffuse faster through a porous solid?

a) Gas molecules.

Gas molecules would diffuse faster through porous media as they are more mobile and can move more freely than liquid molecules.

Fick's first law

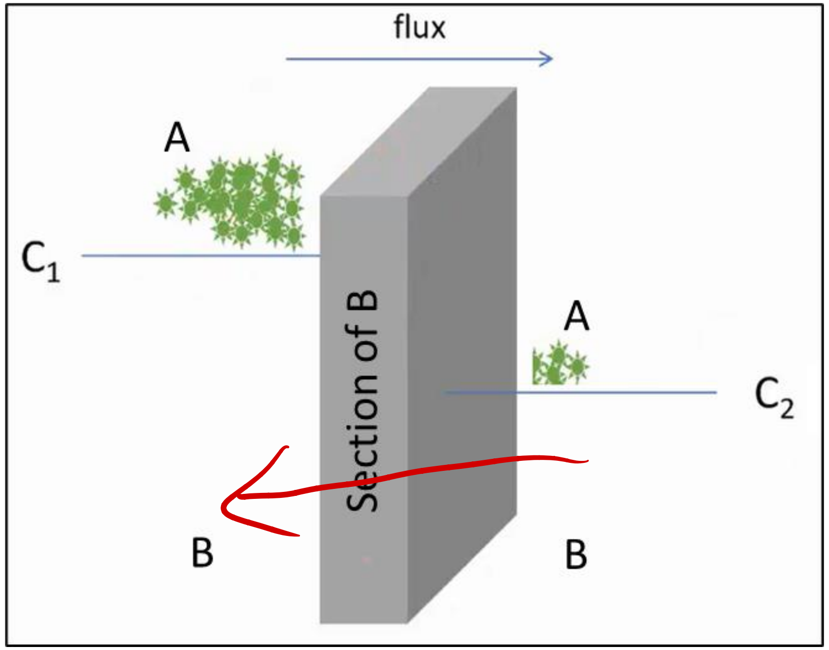
If the diffusion of a species A into a medium B occurs only along the x-direction, the first Fick's law states that the flux J of the species A through B is given by:

Fick's first law

$$\boxed{J = -D_{AB}\frac{dC}{dx}}$$

D_{AB}: diffusion coefficient, which depends on the couple A and B. Units [$m^2\ s^{-1}$].
C: concentration of A.

The negative sign indicates that the flux has opposite direction to the concentration gradient.



Fick's first law and steady state conditions

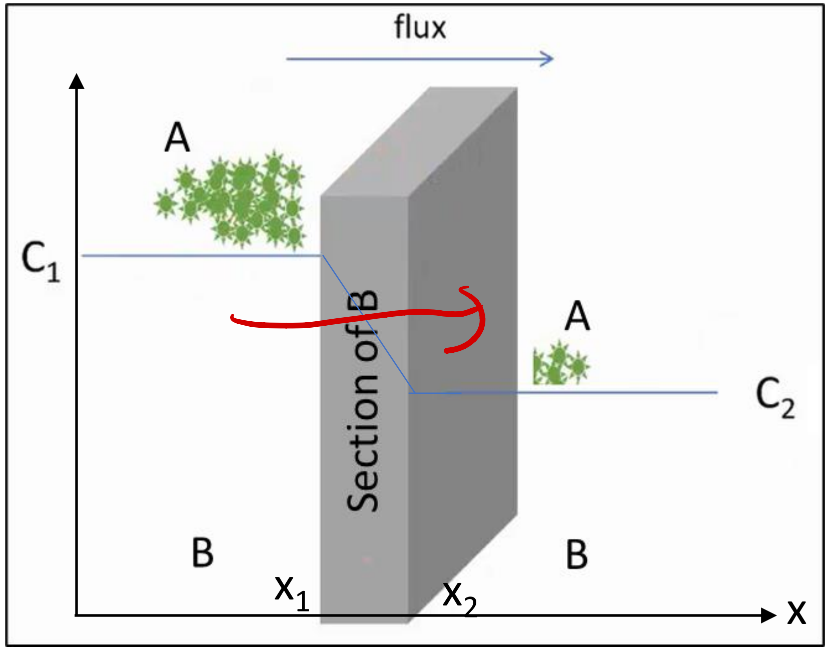
In steady state conditions, the concentration profile and the diffusion flux are constant over time.

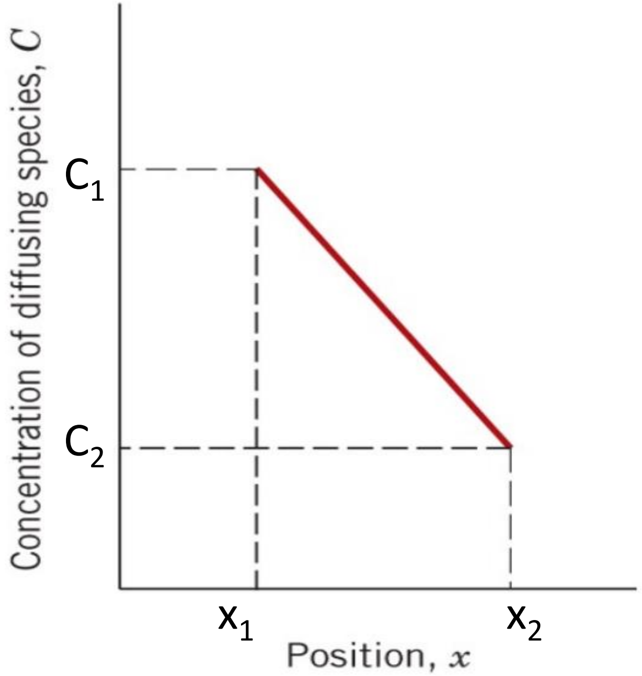
The mass of diffusing species entering the plate on one side is equal to the mass exiting from the other side.

In specific cases, **a linear concentration gradient** is realized, in which case the Fick's law can be written as:

$$J$$

$$\boxed{J = -D_{AB}\frac{\Delta C}{\Delta x} = -D_{AB}\,\frac{C_2 - C_1}{x_2 - x_1}}$$





Poll – Answers in WeChat

Q1) If equal amounts of He and Ar are placed in a porous container and are allowed to escape, which one would escape faster? And how much faster?

a) He
b) Ar

He r↓

Ne

Ar

Poll – Answer

Q1) If equal amounts of He and Ar are placed in a porous container and are allowed to escape, which one would escape faster? And how much faster?

a) He

He escapes faster since it has a smaller atomic weight than Ar. It does so 3.16 times faster due to difference in atomic radius.

?

Applications of Fick's first law

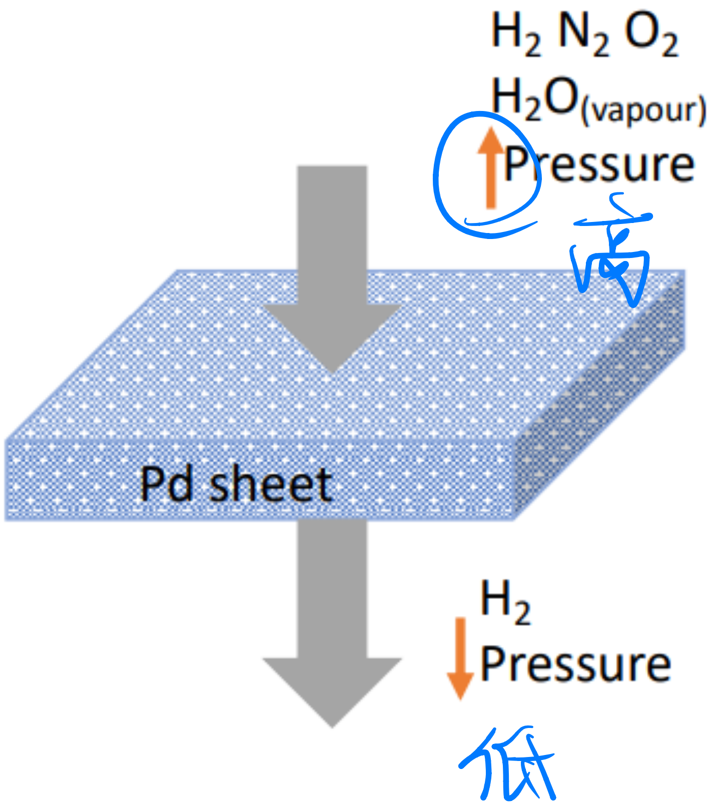
<u>Example 1:</u> Determination of diffusion flux during purification of H_2 using Pd sheet.

Calculate the number of kilograms of hydrogen that pass per hour through a 6-mm thick sheet of palladium having an area of 0.25 m^2 at 600°C.

D

Assume a diffusion coefficient of 1.7×10^{-8} m^2/s, and the respective concentrations at the high- and low-pressure sides of the plate are 2.0 and 0.4 kg of H_2 per m^3 of Pd, and that steady-state conditions have been attained.

$$J = -1.7\times10^{-8}\,\frac{0.4-2}{6\times10^{-3}}$$



Applications of Fick's first law

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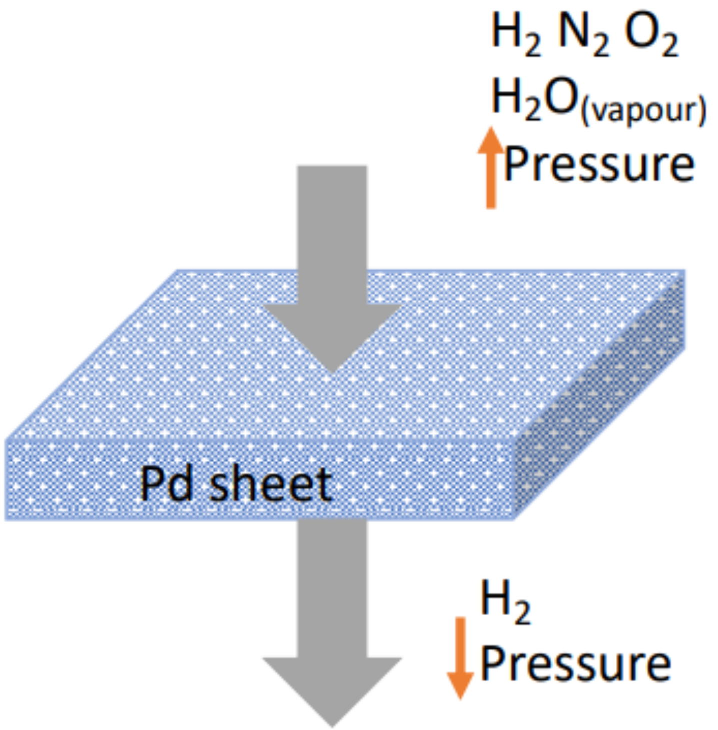

单位面积

$$J = -D\frac{dC}{dx}$$

$$\frac{dC}{dx} = \frac{0.4 - 2}{0.006} = -266.7kg/m^4$$

$$J = -D\frac{dC}{dx} = -1.7 \times 10^{-8}\text{m}^2/s \times \left(-\frac{266.7kg}{m^4}\right) = 4.5339 \times 10^{-6}kg/(m^2s)$$

$$M(kg/h) = 0.25m^2 \times \frac{3600s}{h} \times 4.5339 \times \frac{10^{-6}kg}{m^2s} = 4.08051 \times 10^{-3}kg/h$$

J×面积

换算单位

Applications of Fick's first law

Example 2: Determination of diffusion flux of C in Fe.

A plate of iron is exposed to a carbon-rich atmosphere on one side and carbon-deficient atmosphere on the other side at 700°C.

If a condition of steady state is achieved, calculate the diffusion flux of carbon through the plate if the concentrations of carbon at positions of 5 and 10 mm beneath the carbon-deficient surface are 1.2 and 0.8 kg/m^3, respectively. Assume a diffusion coefficient of 3×10^{-11} m^2/s.

Applications of Fick's first law

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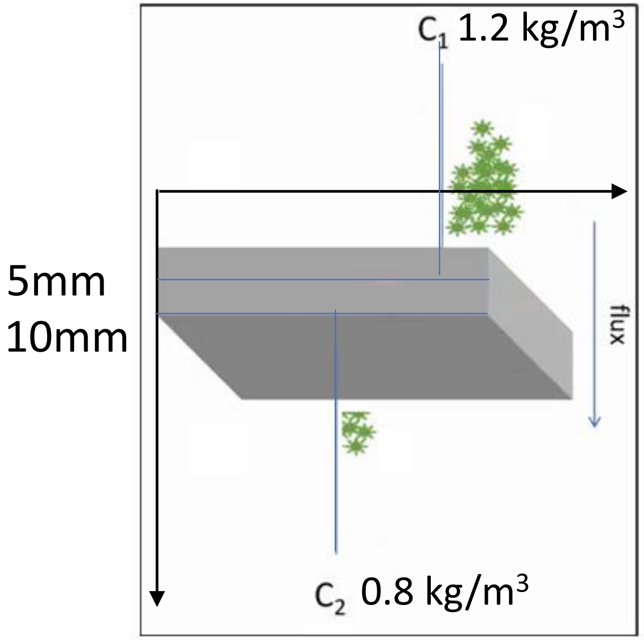

$$\boxed{J = -D\frac{dC}{dx}}$$

$$\boxed{\frac{dC}{dx} = \frac{0.8 - 1.2}{0.005} = -80kg/m^4}$$

$$\boxed{J = -D\frac{dC}{dx} = -3 \times 10^{-11}\mathrm{m}^2/\mathrm{s} \times \left(-\frac{80kg}{m^4}\right) = 240 \times 10^{-11} kg/(m^2 s)}$$

Applications of Fick's first law

<u>Example 3:</u> Determination of concentration of nitrogen diffusing in steel.

A sheet of steel 2.5-mm thick has nitrogen atmospheres on both sides at 900°C and is permitted to achieve a steady-state diffusion condition. The diffusion coefficient for nitrogen in steel at this temperature is 1.85×10^{-10} m^2/s, and the diffusion flux is found to be 1.0×10^{-7} kg/m^2s. Also, it is known that the concentration of nitrogen in the steel at the high-pressure surface is 2 kg/m^3.

How far into the sheet from this high-pressure side will the concentration be 0.5 kg/m^3? Assume a linear concentration profile.

Applications of Fick's first law

<u>Example 3 solution:</u> Determination of concentration of nitrogen diffusing in steel.

A sheet of steel 2.5-mm thick has nitrogen atmospheres on both sides at 900°C and is permitted to achieve a steady-state diffusion condition. The diffusion coefficient for nitrogen in steel at this temperature is 1.85×10^{-10} m²/s, and the diffusion flux is found to be 1.0×10^{-7} kg/m²s. Also, it is known that the concentration of nitrogen in the steel at the high-pressure surface is 2 kg/m³.

How far into the sheet from this high-pressure side will the concentration be 0.5 kg/m³? Assume a linear concentration profile.

$$\boxed{J = -D\frac{dC}{dx}}$$

$$\boxed{1.0 \times 10^{-7} = -1.85 \times 10^{-10}\frac{-1.5}{\Delta x} = \frac{2.775 \times 10^{-10}}{\Delta x}}$$

$$\boxed{\Delta \mathrm{x} = \frac{2.775 \times 10^{-10}}{1.0 \times 10^{-7}} = 2.775mm}$$

Summary

- Transfer of mass within 2 components allows the mixing of one component into another one, this process is known as diffusion;
- Diffusion through crystal structures occur because there are vacant atomic spaces;
- The direction of the diffusion process is determined by the crystal structure of the compound;
- Fick's first law can be used to describe the diffusion flux in a variety of solids in steady state conditions.